Equivalence Partitioning (cont.)

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Equivalence Partitioning : Defect Masking

- Certain defects might possibly remain undetected if different invalid equivalence classes are combined in the same test case!

- Example
  - Input Value 1 <= value <= 99; colour IN (red, green, yellow)
- Equivalence Classes
  - value_VP1: 1 <= value <= 99
  - value_IP1: value < 1
  - value_IP2: value > 99
  - colour_VP1: colour IN (red, green, yellow) e.g. red
  - colour_IP1: NOT colour IN (red, green, yellow) e.g blue
- Test data value=0, colour=blue -> value_IP1 and colour_IP1
- ⇒ Any defective treatment of colour=blue maybe remain undetected

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EP: Analyzing the specification

- A piece of code computes the price of a product based on its value, a discount in % and shipping costs (RM6, RM9 or RM12, depending on shipping mode)
- **Requirements:**
  - ➲Value of goods is given as a positive number with 2 decimal places
  - ➲Discount is a percentage value without decimal places between 0% and 100%
  - ➲Shipping costs can only be 6, 9 or 12

➲Derive the EP class and representative values for each class.

Variable	Equivalence Class	Status	Representative
Value of goods	EC_{11}: x >= 0.00	valid	1000.00
	EC_{12}: x < 0.00	invalid	–1000.00
	EC_{13}: x non-numerical value	invalid	fred
Discount	EC_{21}: $0\% \leq x \leq 100\%$	valid	10%
	EC_{22}: x < 0%	invalid	–10%
	EC_{23}: x > 100%	invalid	200%
	EC_{24}: x non-numerical value	invalid	fred
Shipping costs	EC_{31}: x = 6	valid	6
	EC_{32}: x = 9	valid	9
	EC_{33}: x = 12	valid	12
	EC_{34}: $x \neq \{6, 9, 12\}$	invalid	4
	EC_{35}: x non-numerical value	invalid	fred

- **All test cases:**
  - 10 test cases are derived: 3 positive (valid values) and 7 negative (invalid values) test cases:

Variable	Status	Representative	T01	T02	T03	T04	T05	T06	T07	T08	T09	T10
Value of goods	valid	1000	X	X	X	X	X			X	X	X
	invalid	–1000.00						X				
	invalid	fred							X			
Discount	valid	10%	X	X	X	X	X	X	X			
	invalid	–10%								X		
	invalid	200%									X	
	invalid	fred										X
Shipping costs	valid	6	X					X	X	X	X	X
	valid	9		X								
	valid	12			X							
	invalid	4				X						
	invalid	fred					X					
Expected Output												
Price			906									
Error Message						"Invalid Shipping costs"	"Invalid Shipping costs"					
Actual Output												
Price												
Error Message												
Pass / Fail												

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- **Test cases for valid EC:**
  - Valid equivalence classes provide the following combinations or test cases: T01, T02 and T03

Variable	Equivalence Class	Status	Representative	T01	T02	T03
Value of goods	EC_{11}: x >= 0	valid	1000.00			
	EC_{12}: x < 0	invalid	–1000.00			
	EC_{13}: x non-numerical value	invalid	fred			
Discount	EC_{21}: $0\% \leq x \leq 100\%$	valid	10%			
	EC_{22}: x < 0%	invalid	–10%			
	EC_{23}: x > 100%	invalid	200%			
	EC_{24}: x non-numerical value	invalid	fred			
Shipping costs	EC_{31}: x = 6	valid	6			
	EC_{32}: x = 9	valid	9			
	EC_{33}: x = 12	valid	12			
	EC_{34}: $x \neq \{6, 9, 12\}$	invalid	4			
	EC_{35}: x non-numerical value	invalid	fred			

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- **Test cases for invalid EC:**
  - The following test cases were created using the invalid EC, each in combination with valid ECs of other elements:

Variable	Equivalence Class	Status	Representative	T04	T05	T06	T07	T08	T09	T10
Value of goods	EC_{11}: x >= 0	valid	1000,00							
	EC_{12}: x < 0	invalid	–1000,00							
	EC_{13}: x non-numerical value	invalid	fred							
Discount	EC_{21}: 0% ≤ x ≤ 100%	valid	10%							
	EC_{22}: x < 0%	invalid	–10%							
	EC_{23}: x > 100%	invalid	200%							
	EC_{24}: x non-numerical value	invalid	fred							
Shipping costs	EC_{31}: x = 6	valid	6							
	EC_{32}: x = 9	valid	9							
	EC_{33}: x = 12	valid	12							
	EC_{34}: x ≠ {6, 9, 12}	invalid	4							
	EC_{35}: x non-numerical value	invalid	fred							

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Equivalence Partitioning – in general /1

- Partitioning
  - –The quality of the test depends on precisely segmented variables/elements in equivalence classes
  - –EC that were not identified hold the risk of overlooking possible defects, since the representatives used did not cover all possibilities
- Test cases
  - –Equivalence class method provides test cases for which a representative still has to be chosen
  - –Test data combinations are selected by defining the representative or representatives of each equivalence class

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EP – In general (2)

- choosing representatives
  - **any value** within the EC can be a **representative**. Optimal are:
    - typical values (used often)
    - problem values (suspected failures)
    - boundary values (on the edge of the EC)

➲Representatives of **valid EC** may be **combined**

➲Representatives of **invalid EC** may **not be combined**

➲Representatives of **invalid EC** may only be **combined with valid** representatives of **other EC**

➲For test cases, representatives of invalid EC should be combined with always the same values of other valid EC (standard combinations)

➲Choosing representatives implies that the function within the program uses compare operations

EP– coverage

- Equivalence class coverage can be used as exit criteria to end testing activities

$$EC\text{-}Coverage = \frac{Number\ of\ EC\ tested}{Number\ of\ EC\ defined} * 100\%$$

- Example:
  - If 18 equivalence classes have been determined from the statements or specifications of the input date, and if only 15 of these have been tested in test cases, an equivalence class coverage of 83 % has been reached.
  - Equivalence class coverage = (15 / 18) * 100% = 83,33 %